

Allyl Glycidyl Ether-Based Polymer Electrolytes for Room Temperature Lithium Batteries

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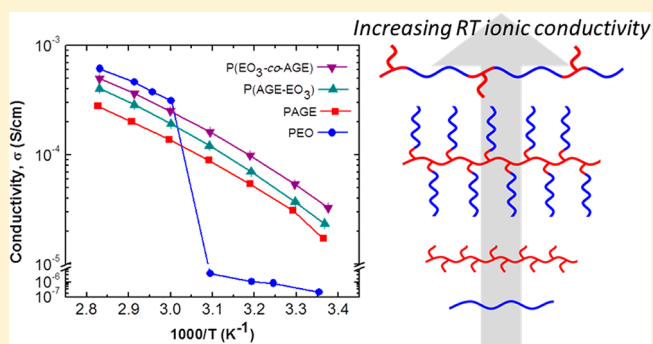
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Supporting Information

ABSTRACT: A new family of polymer electrolytes based on a poly(allyl glycidyl ether) (PAGE) platform has been developed that overcomes many of the limitations of poly(ethylene oxide) (PEO) for battery electrolyte applications. PAGE was shown to have peak conductivities at $[O]/[Li] = 16$, with $\sigma > 3 \times 10^{-5}$ S/cm at 25 °C and $\sigma > 5 \times 10^{-4}$ S/cm at 80 °C. Below 60 °C, PAGE has a conductivity that is 10–100 times higher than that of PEO at equivalent salt concentrations with this disparity in conductivities between PAGE and PEO increasing with decreasing temperature. In addition, the synthetic versatility of allyl glycidyl ether as a building block is demonstrated by the preparation and evaluation of various AGE–EO macromolecular architectures that show superior performance to both PAGE and PEO.



INTRODUCTION

As our energy infrastructure begins to rely on renewable but more intermittent energy sources such as solar and wind, safe, high capacity energy storage devices will become increasingly important. Polymer electrolytes hold promise for the development of lithium batteries with increased energy density that can be adapted to a variety of applications. By eliminating the need for volatile, flammable, and toxic small-molecule electrolytes, these solvent-free, solid-state batteries, in which a moderate to high molecular weight polymer is mixed with a lithium salt, also increase battery safety and introduce new degrees of design flexibility.¹ An additional motivation behind the design of solid polymer electrolytes is that highly energetic metallic lithium anodes may be safely used in lieu of less-energetic intercalation compound anodes.²

In examining the general area of battery materials, a major challenge to advancing these systems commercially has been the development of a polymer material that is able to dissolve and dissociate lithium salts and allow for high lithium ion mobility while simultaneously maintaining mechanical strength and electrode separation. To date, poly(ethylene oxide) (PEO) has been the most frequently and thoroughly studied polymer electrolyte due to its effective solvating properties and ionic conductivities greater than 10^{-4} S/cm above 70 °C.^{3,4} However, below 65 °C, ionic conductivity in PEO decreases dramatically as a result of crystallization, thus making PEO ill-

suited for most battery applications that require operation at ambient temperatures.⁵ As a result, significant recent work has sought to modify PEO to eliminate crystallinity while retaining enhanced solvating, conducting, and structural properties. Strategies employed include PEO oligomer cross-linking,^{6–11} synthesis of rubbery block copolymers^{12–14} with or without incorporation of plasticizing agents,¹⁵ and, most successfully, development of branched and/or supramolecular architectures such as stars/combs^{16–18} and dendritic structures.^{19–21} Other approaches include gel polymer electrolytes and composite materials.^{3,22–25} While these techniques have yielded increases in conductivity at lower temperatures, as yet they have been unable to achieve the high conductivities deemed necessary for commercial viability.²⁶ Moreover, continuing improvements have come at the cost of increasingly sophisticated synthetic schemes that might not be feasible on an industrial scale.

As a result, a polymer electrolyte that is inexpensive, straightforward to produce, able to dissolve lithium salts, and promotes high levels of lithium ion conduction at ambient temperatures is critical to wide-scale commercialization of lithium–polymer batteries. Thus, we propose as an alternative to PEO a family of polymer electrolytes that are amorphous

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over all useful temperatures, can be synthesized in one or two steps, have improved ability to conduct ions at room temperature, and should be easily adapted into existing solid polymer electrolyte schemes in a range of battery devices.¹³ Poly(glycidyl ether)s (PGEs) potentially offer these desirable traits as they are synthetically similar to PEO, but they contain pendant ether groups that inhibit crystallization.²⁷

An ideal polymer electrolyte should contain a large fraction of heteroatoms such as oxygen to promote salt dissociation coupled with a low glass transition temperature indicative of increased segment mobility and ion transport. In evaluating poly(glycidyl ether)s as a broad platform for polymer electrolytes, we selected poly(allyl glycidyl ether) (PAGE) as a matrix material due to its low T_g near -80 °C and high oxygen content. Prior studies have only investigated the use of AGE units as cross-linking functionalities in statistical or block copolymers with ethylene oxide.^{6,7,9–11} In these instances, the amount of AGE incorporated in the polymer was typically less than 10 mol %, and the dangling allylic functionality was eliminated through cross-linking. In another case, copolymers with AGE had high PDIs and were not well-defined.²⁸ As a result, these studies overlook the significant potential of PAGE homopolymer and related copolymers as synthetically accessible, versatile polymer electrolytes. One advantage of these PAGE systems is the amorphous nature, suggesting that a PAGE-based electrolyte could be utilized in a room temperature battery in place of PEO. In addition, the chemistry of PAGE allows for secondary functionalization, high molecular weights, and low PDIs,²⁹ making PAGE a model system to explore the potential of poly(glycidyl ether) derivatives as polymer electrolytes for lithium–polymer batteries.

RESULTS AND DISCUSSION

Poly(allyl glycidyl ether) was synthesized by anionic ring-opening polymerization of allyl glycidyl ether utilizing benzyl alcohol deprotonated by potassium naphthalenide as the initiating system.²⁹ Polymerization in the melt at 30 °C over 20 h yielded PAGE with controllable molecular weights and PDIs of 1.15 or less. 14 kg/mol PAGE was blended with bis(trifluoromethylsulfonimide) (LiTFSI), a standard lithium salt with a bulky anion that has typically shown good performance in polyether-based electrolytes.³⁰ As the conductivity of polymer electrolytes is known to be highly dependent on salt concentration, PAGE was combined with LiTFSI at a range of concentrations from $r = 0.01$ to $r = 0.10$, where r is the molar ratio of lithium in the salt to oxygen heteroatoms in the polymer, $[Li]/[O]$. The conductivities of the PAGE–LiTFSI complexes are shown in Figure 1 as a function of temperature for several LiTFSI concentrations.

The conductivity of the pure, undoped PAGE material was measured since a desirable feature of any polymer electrolyte is that its electrical conductivity is low, while ionic conductivity is high.²⁶ Over the entire temperature range investigated (25–80 °C), the ionic conductivity of pure PAGE is extremely low, $\sigma < 10^{-8}$ S/cm, assuring that any salts remaining from the synthesis contribute little to the measured ionic conductivity. Significantly, on the addition of a small amount of LiTFSI ($r = 0.01$), the ionic conductivity increases approximately 3 orders of magnitude, demonstrating the facile transport properties of dissociated lithium salts in PAGE. The ionic conductivity of PAGE continues to increase with increasing salt concentration up to a loading of $[O]/[Li] = 10$, or approximately 33 wt % LiTFSI, where the conductivity reaches a maximum with

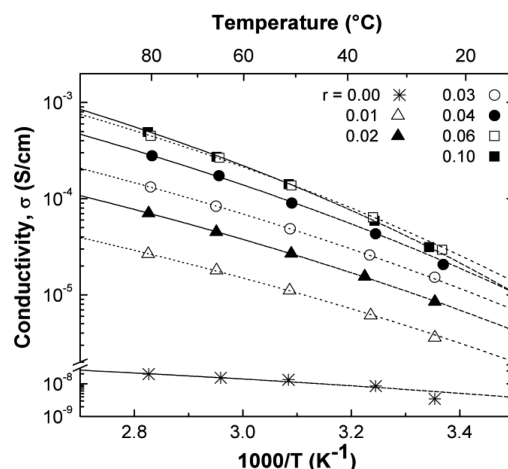


Figure 1. Average ionic conductivity (σ , S/cm) of PAGE–LiTFSI complexes versus temperature (top) and $1000/T$ (bottom) for PAGE–LiTFSI complexes for a range of lithium ion concentrations. Error bars shown are derived from the standard deviation of multiple temperature sweeps on each sample. Lines are fits of the VTF equation to the data. Relative standard deviation (RSD) for Li-containing samples is less than 6.1% (average 1.6%). RSD for $r = 0.0$ is less than 51% (average 17%).

respect to LiTFSI concentration. Maximum mobility and conductivity are achieved in the range $r = 0.06$ – 0.10 , where the conductivity appears to plateau. In many polymer electrolytes conductivity increases with increasing salt incorporation, but at higher concentrations, conductivity will decay due to reduced salt dissociation and decreasing segmental mobility.^{17,24} Above $r = 0.10$, LiTFSI appeared to exceed the solubility limit in PAGE, resulting in a cloudy appearance, and it is likely that conductivity would eventually decrease.

Traditionally, it is believed that the lithium ions are solvated by coordination to oxygen heteroatoms in the backbone of polyether systems, and this coordination has consequences for the segmental mobility of the polymer. For PAGE, T_g was observed to increase linearly with LiTFSI concentration as shown in Figure 2. At the highest loading, $r = 0.10$, T_g has increased from -78 to -49 °C, suggesting that the coordinated lithium ions are functioning as transient cross-links by complexing with two or more chains.^{10,31}

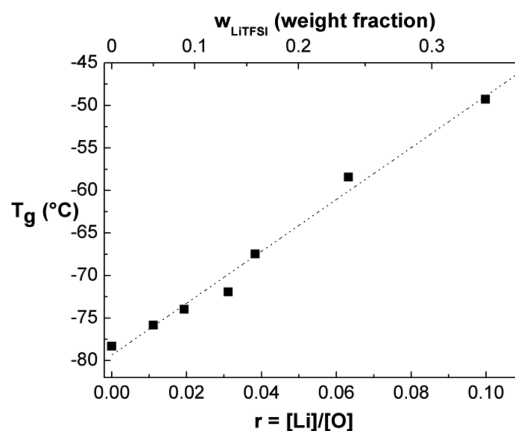


Figure 2. T_g as determined by DSC vs lithium concentration $r = [Li]/[O]$. On the top axis, weight fraction of LiTFSI in electrolyte is given. T_g increases linearly with r . The dashed line is a linear fit to the data.

To better understand the performance of these materials, the temperature dependence of ionic conductivity over all salt concentrations was examined and found to be well described by the Vogel–Tammann–Fulcher (VTF) equation

$$\sigma(T) = \sigma_0 e^{-E_a/R(T-T_0)} \quad (1)$$

in which σ_0 is related to the maximum number of charge carriers, E_a is the limiting high temperature activation energy of ion-conduction/segmental motion, and T_0 is the VTF temperature or equilibrium glass transition temperature, chosen here to be 50 K below the glass transition temperature of the sample.³² Fits to eq 1 are shown in Figure 1 as dashed or dotted lines for each data series, and the corresponding parameters are also given in Table 1. As expected, with increasing addition of

Table 1. Glass Transition Temperatures and VTF Parameters for Polymer Electrolytes

[O]/[Li]	$r = [\text{Li}]/[\text{O}]$	w_{LiTFSI}	T_g^a (°C)	E_a (kJ/mol)	σ_0 (S/cm)
[Li] = 0	0	0.00	−78.0	5.96	6.24×10^{-7}
90	0.01	0.05	−75.8	9.02	5.11×10^{-3}
52	0.02	0.09	−74.0	9.58	1.97×10^{-2}
32	0.03	0.13	−72.0	9.82	4.54×10^{-2}
26	0.04	0.16	−67.5	10.4	1.58×10^{-1}
16	0.06	0.24	−58.4	9.8	2.33×10^{-1}
10	0.10	0.33	−49.3	9.47	2.79×10^{-1}

^aMeasured by DSC. T_0 set to $T_g - 50$ K in VTF fits to conductivity data.

salt, the prefactor σ_0 increases, indicating that the number of charge carriers increases and that the added salt continues to dissociate at high concentrations. The limiting high temperature apparent activation energy, E_a , reaches a maximum at $r = 0.04$ ($[\text{O}]/[\text{Li}] = 26$), implying that above this salt concentration there may be a change in the mode, or energetics of ion conduction, although the differences between E_a may be within error. This is supported by the fit values of σ_0 (assumed to be proportional to the charge carrier density). From these results it can be observed that, on a per mole Li basis, the maximum charge carriers per mole of Li plateaus at about $[\text{O}]/[\text{Li}] = 26$.

Comparison with the ionic conductivity of PEO with PAGE as a function of salt concentration is complicated by the presence of 50% of the oxygen atoms in PAGE being in the allyl ether side chain; hence, not all oxygen atoms are located in the backbone as for PEO. However, the observation that the peak ionic conductivity of PAGE occurs near $r = 0.06$ ($[\text{O}]/[\text{Li}] = 16$), which is approximately the same as that for PEO,¹³ suggests that the side chain does play a role in ion transport. Infrared (FT-IR) spectroscopy was therefore used to investigate the interaction between PAGE and the LiTFSI salt, with a comparison of FT-IR spectra for pure PAGE, LiTFSI, and several concentrations of LiTFSI in PAGE shown in Figure 3.

Infrared studies clearly demonstrate that LiTFSI dissociates in the presence of PAGE as indicated by the shifts in vibrational frequencies of the TFSI anion from 810, 773, and 749 cm^{-1} to 787, 761, and 739 cm^{-1} , respectively.³³ Inversion of the relative peak intensities at 1350 and 1320 cm^{-1} and the shifts in the signals at 1200, 1140, and 1160 cm^{-1} to lower wavenumbers are also all consistent with a completely dissociated TFSI anion.³⁴ Although nearly all the vibrations of LiTFSI shift upon solvation in PAGE, changes in the IR spectrum of PAGE itself are less obvious. It is worth noting that stretches at 3079, 1646,

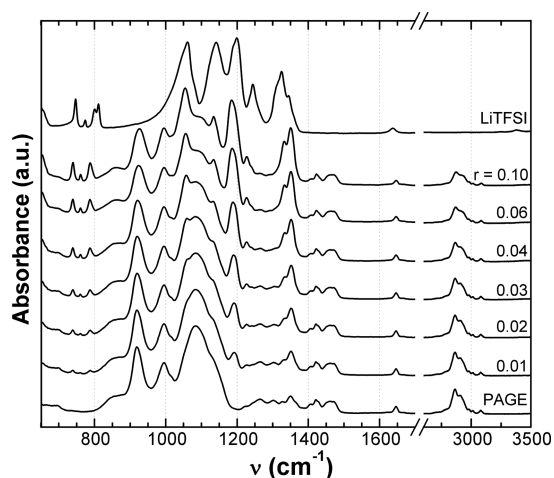


Figure 3. Infrared spectra of PAGE, LiTFSI, and PAGE–LiTFSI complexes at several concentrations: $[\text{O}]/[\text{Li}] = 90, 52, 32, 26, 16$, and 10.

and 995 cm^{-1} , assigned to C–H and C=C alkene vibrations, do not shift even at the highest concentrations of LiTFSI salt. These results suggest little association between LiTFSI and the C=C double bond along the backbone of PAGE with the LiTFSI interacting strongly with cooperative motion of the backbone and side-chain ether units.

The roles of the alkene and ether functional groups in lithium binding were also investigated by ^{13}C NMR of PAGE–LiTFSI complexes, as shown in Figure 4. Peak assignments are

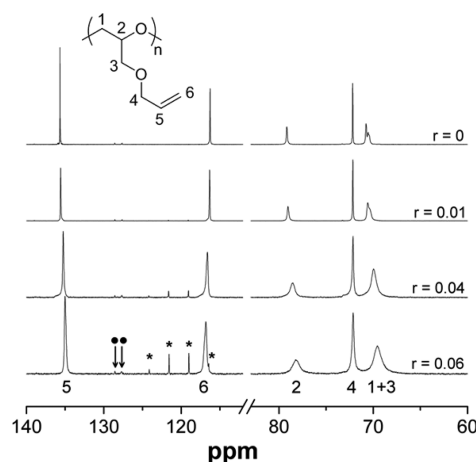


Figure 4. ^{13}C NMR spectra of PAGE (top) and PAGE–LiTFSI mixtures. $[\text{Li}]/[\text{O}]$ ratios are given on the right. Peak assignments are labeled corresponding to the top structure. All carbons shift upfield except C1 which shifts downfield and C4 which does not shift. Solid circles denote peaks arising from the benzylic end group. Asterisks denote appearance of quartet arising from CF_3 in the TFSI anion.

consistent with previous studies^{29,35} and were confirmed via ^{13}C NMR attached-proton-test (APT) experiments with the methylene carbons C1 and C3 overlapping. As LiTFSI is added, the CF_3 quartet of TFSI anion clearly increases in intensity, with the comparison of the salt concentration measured gravimetrically during preparation and by ^{13}C NMR spectroscopy in full agreement. Upon increasing salt concentration, C1–C3 shift upfield and show significant broadening in peak shape, while C4 does not shift at all. C5 likewise shifts upfield,

whereas C6 shifts an equal measure downfield with C4–C6 showing minimal broadening. This suggests that lithium coordination has significantly altered the environment around C1, C2, and C3 which is indicative of strong O–C–C–O coordination with lithium and is typical of polyethers. As suggested by the symmetric shifts of C5 and C6, the solvent environment around the allyl group is also changed by the addition of salt, but the effect is much smaller and we believe the allyl unit does not strongly interact with the LiTFSI, which is in agreement with the infrared studies.

This detailed understanding of the lithium ion solvation by PAGE now allows a direct comparison of the performance of PAGE with PEO of similar molecular weight and at equal weight fractions of LiTFSI incorporation shown in Figure 5.

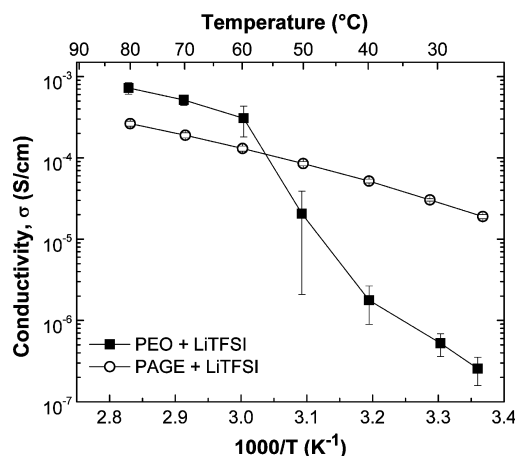


Figure 5. Comparison of the average ionic conductivity of 29 kg/mol PAGE and 20 kg/mol PEO doped with 20 wt % LiTFSI. Error bars shown are derived from the standard deviation of multiple temperature sweeps on each sample.

Above 60 °C, where PEO is completely amorphous, the ionic conduction of PEO is approximately double that of PAGE at equivalent concentration of LiTFSI. However, below 60 °C, PEO conductivity drops significantly due to crystallization while PAGE maintains high conductivities and remains amorphous. The large degree of error seen in the conductivity of PEO between 40 and 60 °C is due to the influence of the degree of crystallization, which is strongly time-dependent. At room temperature, the difference in conductivity between PAGE and PEO exceeds 2 orders of magnitude (100×).

The encouraging performance of PAGE as a polymer electrolyte across a wide temperature range provides strong support for exploiting the synthetic versatility of the allyl units in the design of polymer electrolytes. As a result, two strategies for increasing the oxygen content and for introducing ethylene oxide repeat units within the overall structure were examined. The first involves a graft copolymer strategy where post-functionalization of the allyl side group via thiol–ene coupling allows for the introduction of short ethylene oxide oligomers. For that reason thiol-functionalized monomethoxy dimer and trimer of ethylene oxide were designed and synthesized. As shown in Scheme 1, photochemical coupling of thio-functionalized monomethyl di(ethylene glycol) and tri(ethylene glycol) with PAGE homopolymer proved to be a facile process with complete consumption of the allyl units within 1 h as observed by NMR spectroscopy (spectra provided in the Supporting Information). Gel permeation chromatography also showed the

Scheme 1. Thiolene Coupling of mPEG₂SH and mPEG₃SH to PAGE To Produce PAGE-(EO)₂ and PAGE-(EO)₃

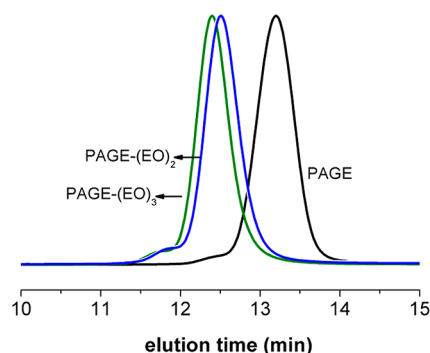
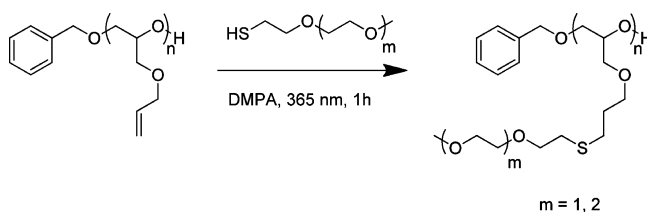


Figure 6. Gel permeation chromatographs of thiolene products PAGE-(EO)₂ and PAGE-(EO)₃ compared to their parent polymer, PAGE.

expected increase in molecular weight with no observable cross-linking reactions (Figure 6). Significantly, no crystallinity was observed for either of these graft systems, and only marginal changes in the glass transition temperature, T_g , from −78 °C to −71 and −75 °C for the dimer- and trimer-functionalized PAGE, respectively, were observed. Unlike previous studies of grafted EO polymers that observed a decrease in T_g with increasing length of the EO side chain,^{17,18} using a PAGE backbone, which has a lower T_g than PEO, leads to an increase in T_g upon grafting EO oligomers. It is a bit surprising that the T_g increase was greater for the dimer. This is likely a side effect of slightly greater amounts of mild cross-linking during the thiolene reaction compared to the trimer.

Ionic conductivities of PAGE, PAGE-(EO)₂, and PAGE-(EO)₃ were measured at a concentration of 20 wt % LiTFSI and are shown in Figure 7 as a function of temperature. The shorter side chain derivative, PAGE-(EO)₂, exhibits enhanced conductivity when compared to PAGE near room temperature with the magnitude of this increase decreasing at higher temperatures. In contrast, the tri(ethylene glycol) derivative, PAGE-(EO)₃, shows an even greater increase in performance at all temperatures which clearly demonstrates the advantages of side-chain modification of PAGE in the design of polyether derivatives with increased conductivity. A previous study of EO oligomers grafted on polysiloxane saw similar increases and a maximum conductivity when the side chain is six EO units.¹⁷ It is possible further increases might be achieved with longer side chains on PAGE as well. However, compared to polysiloxane, the coordinating ability of the PAGE backbone and its existing ether linkages seems to lead to less drastic improvements upon grafting.

An alternative strategy to the graft copolymer approach described above is to investigate copolymerization of AGE with EO to give linear materials in which the introduction of AGE units disrupts crystallinity and also provides functionalization sites along the backbone. In addition, copolymerization of allyl

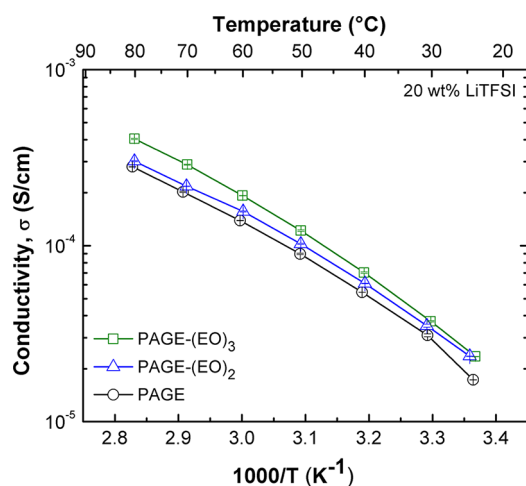
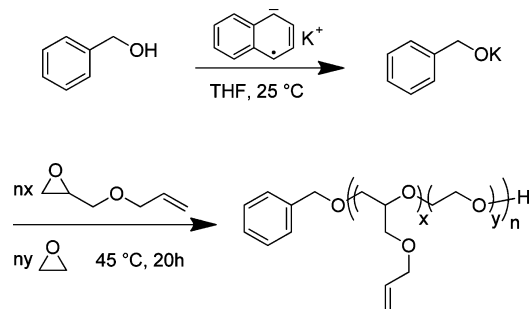


Figure 7. Average ionic conductivity of PAGE versus the thiolene product doped with 20 wt % LiTFSI. Error bars shown are derived from the standard deviation of multiple temperature sweeps on each sample.

glycidyl ether with ethylene oxide is a one-step synthetic strategy that allows for large scale samples to be prepared. While prior copolymerization studies of EO and AGE have been limited to low levels of AGE incorporation¹⁰ and afforded poorly defined materials,²⁸ recent work from our group has demonstrated the full range of AGE/EO copolymerization ratios leading to well-defined P(EO-co-AGE) materials with low PDIs and controlled molecular weights (Scheme 2).³⁶

Scheme 2. Synthesis of Well-Defined P(EO-co-AGE)



In this study, copolymers were synthesized from feed ratios of [EO]:[AGE] = 1 to [EO]:[AGE] = 7.5 to give copolymers with weight ratios of EO ranging from 32 to 77 wt %. Detailed characterization data for each P(EO-co-AGE) material are found in Table 2. Significantly, DSC analysis of these materials showed that the level of crystallinity and any associated melting transitions could be accurately tuned by changing the weight

Table 2. P(EO-co-AGE) Characterization and Properties

molar feed ratio [EO]/[AGE]	polymer % EO (wt)	M_w (kg/mol) by NMR	M_n (kg/mol) by GPC	PDI	T_g (°C)	X_c (%)
PEO	100	n. a.	33.7	1.07	n.a.	86
7.5	77	39.4	29.5	1.12	−69	32
5	69	16.9	14.3	1.21	−70	23
4	61	14.7	15.8	1.13	−71	15
3	56	6.5	7.9	1.13	−69	0
1	32	27.5	17.3	1.11	−73	0

ratio of EO in the copolymer. This insertion of AGE repeat units in the polymer backbone breaks up the PEO sequences into smaller sections, reducing the melting temperature as well as the overall crystalline fraction of the polymers. As can be seen in Figure 8, no melting transitions were observed up to 61

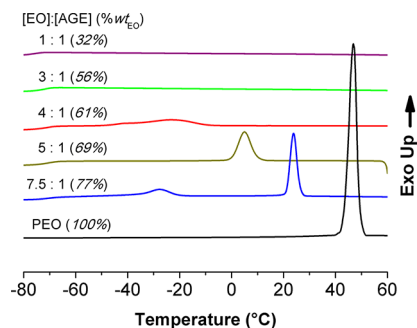


Figure 8. DSC cooling traces of P(EO-co-AGE) polymers at various monomer ratios. Traces are baseline normalized.

wt % of EO with only a minor melting transition at 5 °C for copolymers with 68 wt % EO. Only at the highest incorporation of EO, 77 wt % EO, does significant crystallization occur, and the melting temperature (T_c) is approximately room temperature (25 °C).

This tunability of AGE/EO copolymers now allows a thorough analysis of structure/property relationships and ionic conductivities for the copolymers doped with 20 wt % LiTFSI to be performed. As shown in Figure 9, a systematic

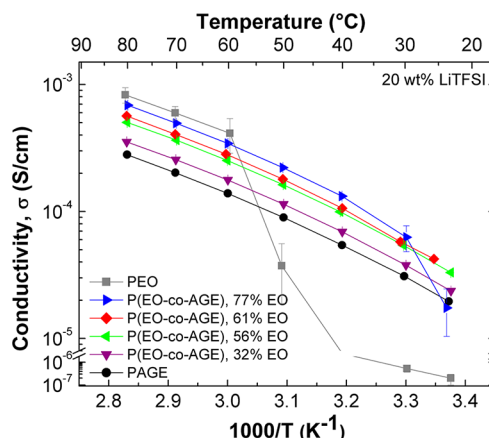


Figure 9. Average ionic conductivity of P(EO-co-AGE) with varying incorporations of EO by weight. Error bars shown are derived from the standard deviation of multiple temperature sweeps on each sample. Greater fraction of EO leads to near linear increases in conductivity. However, this is offset by increasing T_g and appearance of T_c for >32 wt % EO. For 77 wt % EO incorporation, a strong decrease in conductivity is marked between 30 and 25 °C, corresponding to crystallization.

increase in conductivity is observed with increasing EO content. Of particular note is the performance of the 77 wt % EO sample which showed comparable conductivity to PEO at temperatures above 60 °C with markedly improved performance at lower temperatures. At 40 °C, a conductivity of 10^{-4} S/cm for the 77 wt % copolymer is observed which is 5 times higher than for PAGE homopolymer and 3 orders of magnitude higher than for PEO. This dramatic increase in performance is

clearly related to the negligible effect that AGE units have on the conductivity coupled with their ability to disrupt crystallinity while retaining very low glass transition temperatures ($-69\text{ }^{\circ}\text{C}$). In order to demonstrate the retained functionality of the copolymers, cross-linking of the 61 wt % EO polymer could also be achieved, resulting in a significant increase in mechanical stability. However, conductivity was only reduced by less than 5%, suggesting that the chemistry associated with the allyl units can be exploited beyond the ability of the AGE repeat units to disrupt crystallinity. In addition, the copolymers offer overall synthetically simpler routes to well-defined materials that could be produced on a large scale.

The high room temperature conductivities observed in these materials are promising for the development of low temperature polymer electrolyte-based batteries. Development and performance testing of devices will be the subject of future studies. Recent work³⁷ showing high thermal stability and a wide electrochemical window for allyl-containing polymers in the presence of LiTFSI is encouraging for the future incorporation of PAGE-based polymer electrolytes in batteries.

CONCLUSIONS

We have demonstrated that a poly(glycidyl ether), specifically poly(allyl glycidyl ether), is an attractive alternative polymer electrolyte to the ubiquitous poly(ethylene oxide) for a nonaqueous electrolyte layer in batteries that is operable at room temperature. The pendant allyl ether not only inhibits the formation of nonconducting crystalline regions but also aids in ion solvation and conduction. Additionally, PAGE provides additional advantages through the reactive allyl side group, which we have shown can be used to increase the conductivity through the addition of short ethylene oxide oligomers. The highest conductivities were achieved via a copolymer architecture containing EO and AGE. These materials are particularly promising because not only do they boast conductivities near 10^{-4} S/cm at room temperature but also they retain the functionality of PAGE which can be used to cross-link the composition or introduce other useful groups, such as ionomers.

EXPERIMENTAL SECTION

Materials and Sample Preparation. Bis-(trifluoromethylsulfoniimide) (LiTFSI) was purchased from Sigma-Aldrich and dried *in vacuo*. PAGE was synthesized via anionic ring-opening polymerization in accordance with known methods described in detail elsewhere.²⁹ Allyl glycidyl ether (AGE) was purchased from TCI America, degassed by three freeze–pump–thaw cycles, stirred over butyl magnesium chloride, and distilled prior to use. The purified AGE was polymerized in the melt at $40\text{ }^{\circ}\text{C}$ with a potassium benzoxide initiator and terminated with methanol. A radical inhibitor (0.01 wt % BHT) was added to prevent cross-linking. A 14 kg/mol ($\text{PDI} = 1.14$) PAGE was used for ^{13}C NMR studies while 29 kg/mol ($\text{PDI} = 1.10$) PAGE was used for all other investigations. 20 kg/mol PEO was purchased from Sigma-Aldrich and dried *in vacuo* before use. Molecular weights were determined by ^1H NMR, and polydispersity indices were determined by size exclusion chromatography in chloroform relative to polystyrene standards.

Thiol-functionalized monomethoxy dimer and trimer of ethylene oxide (mPEG₂SH and mPEG₃SH) were synthesized from their alcohol analogues in accordance with known methods.^{38,39} When starting from the heterobifunctional oligomers with a methoxy and alcohol end group, the latter was activated by reaction with tosyl chloride in high yield (>90%). A subsequent nucleophilic displacement of the tosyl

group with thioacetate (dimer 83%, trimer 89%) followed by cleavage of the acetate under acidic conditions resulted in pure thiol-functionalized monomethoxy dimer and trimer ethylene oxide in excellent yields (dimer 89%, trimer 99%) after purification with column chromatography. For synthetic details the reader is referred to the Supporting Information.

The thiol–ene reaction of mPEG₂SH and mPEG₃SH with PAGE was undertaken as follows. PAGE was dissolved in benzene (Sigma-Aldrich) at a concentration of 30 wt %. 5–6 equiv of the thiol was added followed by 0.1 equiv of 2,2-dimethoxy-2-phenylacetophenone (DMPA, Sigma-Aldrich). The mixture was sparged with argon for 30 min and then irradiated with 365 nm light in 15 min intervals. Conversion of allyls to thiol–ene product was monitored NMR. All reactions reached 100% conversion within 1 h.

To prepare the polymer electrolytes, the polymers were first dried overnight under high vacuum and then transferred while under vacuum to an argon-filled glovebox where the polymers were mixed with LiTFSI. PAGE–LiTFSI and P(EO-*co*-AGE) mixtures were stirred at $50\text{ }^{\circ}\text{C}$ for at least 3 h. PEO–LiTFSI mixtures were prepared by heating PEO to $80\text{ }^{\circ}\text{C}$ until molten and adding LiTFSI and then stirred at $80\text{ }^{\circ}\text{C}$ for at least 1 h. All polymer electrolyte mixtures were prepared and stored in a dry glovebox ($\text{H}_2\text{O} < 0.1\text{ ppm}$) under an argon atmosphere.

Polymer Electrolyte Characterization. A glass cell containing two platinum electrodes supplied by Topac, Inc., was immersed in the polymer–salt mixture and sealed under argon. Ionic conductivities were then measured by ac impedance spectroscopy performed on a VMP3 potentiostat using EC-Lab software. The complex impedance spectra were collected over the frequency range of 500 kHz–10 mHz. Each sample was equilibrated at the given temperature for at least 1 h prior to measurement, and measurements were collected on both heating and cooling cycles. Measured temperatures are accurate to within $\pm 1\text{ }^{\circ}\text{C}$. The cell constant provided by the manufacturer was confirmed using a 0.01 M potassium chloride solution.

Differential scanning calorimetry thermograms were collected using a TA Instruments Q2000 MDSC at a heating rate of $2\text{ }^{\circ}\text{C/min}$, unless otherwise noted. The glass transition temperature (T_g) was determined as the inflection point on the second heating cycle from -90 to $100\text{ }^{\circ}\text{C}$. FT-IR spectra of PAGE–LiTFSI complexes were acquired on a PerkinElmer ATR-FTIR under ambient conditions. ^{13}C and ^1H NMR spectroscopy was carried out on a Bruker AVANCE500 spectrometer at room temperature. ^{13}C NMR was performed in the melt by inserting a sealed capillary containing D_2O for locking and shimming into neat PAGE–LiTFSI samples.

ASSOCIATED CONTENT

Supporting Information

Experimental details. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

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